FARHAN RAHMAN ARNOB

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EDUCATION

Ph.D. Computer Science and Engineering, University of North Texas, Denton, TX Expected – May 2027
- GPA: 3.85/4.00; Focus: Machine Learning and Software Engineering
- Dissertation research: machine learning for automated bug triage
- Advisor: Dr. Renee Bryce; Committee: Dr. JungHwan Oh, Dr. Xuan Guo, Dr. Wajdi Aljedaani

M.S. Computer Science, University of North Texas, Denton, TX July 2025
- GPA: 3.83/4.00

B.Sc. Computer Science and Engineering, American International University–Bangladesh, Dhaka January 2018
- GPA: 3.63/4.00

TEACHING INTERESTS

Software Engineering; C++; Python; Java programming; Data structures; Software Verification; IT and Software Project Management; Software Development Capstone; Mobile Application Development Programming (Java, Kotlin); Introduction to Artificial Intelligence and Machine Learning; Natural Language Processing; Software Testing and Quality Assurance; Human-Computer Interaction; Database Systems.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Dept. of Computer Science and Engineering, UNT August 2022 – Present

Courses supported:
- **CSCE 3055 IT Project Management** Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026, Fall 2026
- **CSCE 5270 Computer-Human Interfaces** (graduate) Summer 2023
- **CSCE 3530 Introduction to Computer Networks** Summer 2024
- **CSCE 5210 Fundamentals of Artificial Intelligence** (graduate) Summer 2025
- **CSCE 3444 Software Engineering** Summer 2026

Key responsibilities and contributions:
- Retained in IT Project Management every fall and spring semester since Fall 2022; 50–100 students per semester in the course that leads directly into the senior capstone.
- Help create the syllabus, assignments, and quizzes; manage the Canvas course; grade; and monitor student progress.
- Mentor students closely for their future job roles; teach and encourage leadership, teamwork, and project planning.
- Supervising instructors: Hadiseh Gooranorimi, Diana Rabah, Dr. Ryan Garlick, and Dr. Moawia Eldow.

MENTORING AND OUTREACH

Research Mentor, Summer Camp Program (NSF REU Site), UNT Summers 2023, 2024, 2025
- REU Site "TaMaLe: Testing and Machine Learning for Context-Driven Systems" (NSF Award #2149969; PI: Dr. Renee Bryce).
- Mentored undergraduate researchers in context-aware systems and machine learning; the 2025 work received the Best Paper Award (Reviewers' Choice) at IEEE CCWC 2026.

Student Mentor, CSCE 3055 IT Project Management, UNT August 2022 – Present

- Coach capstone-bound student teams on planning, leadership, and preparation for industry roles.

Judge, Eagle Awards, University of North Texas March 2025

Judge, TAMS Science Fair, Texas Academy of Mathematics and Science January 2024

Team Development Sessions, Field Information Solutions Ltd., Dhaka 2018 – 2021

- Led sessions for the development team on design patterns, clean code, reactive programming, and responsive Material UI.

RESEARCH INTERESTS

Machine learning for software engineering: automated bug triage and routing, transformer-based natural language processing, learning to rank, and mining software repositories; mobile and context-aware systems and software testing.

RESEARCH EXPERIENCE

Doctoral Researcher, University of North Texas (advisor: Dr. Renee Bryce) January 2023 – Present

- Designed an encoder-agnostic hybrid transformer (RoBERTa, CodeBERT, DeBERTa) that combines bug-report text with seven categorical and nine numerical metadata features; achieved 69.02% Top-1 and 90.69% Top-10 accuracy on 25,220 Mozilla Firefox bug reports across 285 developers, a 15.18-percentage-point Top-1 gain over a text-only baseline (SEKE 2026).
- Introduced training-only commit-message augmentation that keeps models deployable for newly filed bugs; an earlier XLNet model with commit messages raised Firefox Top-1 accuracy from 23.28% to 69.45% and reached 90.89% Top-10 accuracy on Eclipse (ITNG 2025).
- Built a routing-aware learning-to-rank framework (LightGBM LambdaMART) with an implicit routing graph and strictly chronological evaluation on Firefox, Chromium, and Eclipse; Top-1 accuracy of 75.5%, 47.7%, and 65.4% versus 42.0%, 22.9%, and 43.1% for a tossing-graph baseline (under review).
- Prepared the proposal documents for the TACC Lonestar6 computing allocation awarded to PI Dr. Renee Bryce; sole research user running large-scale transformer experiments.
- Extended and rebuilt the lab's mobile application and server system to collect smartphone context data; applied POERMH and TKE-Rules sequence rule mining to predict frequent context-event sequences and improve the efficiency of testing context-sensitive mobile apps.

PUBLICATIONS

Peer-reviewed conference papers

- **F. R. Arnob**, P. Goyal, M. K. Khan, and R. Bryce, "Enhancing Bug Triaging with Organizational Metadata and Commit-Aware Transformers," in *Proc. 38th Int. Conf. on Software Engineering and Knowledge Engineering (SEKE 2026)*, Toronto, Canada, Oct. 2026, doi: 10.18293/SEKE2026-010. Regular paper; 41% acceptance rate.
- P. Goyal, B. Helmholz, T. Tran, **F. R. Arnob**, and R. Bryce, "Hybrid Sequential Pattern Mining and Machine Learning for Context-Aware Event Prediction in IoT," in *Proc. 2026 IEEE 16th Annual Computing and Communication Workshop and Conference (CCWC)*, Las Vegas, NV, USA, Jan. 2026, pp. 917–923, doi: 10.1109/CCWC67433.2026.11393795. **Best Paper Award (Reviewers' Choice).**
- **F. R. Arnob**, R. H. Mollik, P. Goyal, and R. Bryce, "Bug Triaging Based on Transformer Models Utilizing Commit Messages," in *Proc. 22nd Int. Conf. on Information Technology: New Generations (ITNG 2025)*, Springer, 2025, pp. 354–366.

Under review

- "Beyond Next-Hop Bug Tossing: Learning Implicit Routing Patterns for Automated Bug Triage," submitted to the 24th Int. Conf. on Information Technology: New Generations (ITNG 2027); under double-blind review.

Preprints

- M. K. Khan, R. Michaels, **F. R. Arnob**, and R. Bryce, "Prioritization Techniques for Android Test Suites Generated by a Reinforcement Learning Algorithm," SSRN preprint, doi: 10.2139/ssrn.4450321.

PRESENTATIONS

- "Enhancing Bug Triaging with Organizational Metadata and Commit-Aware Transformers," SEKE 2026, October 6, 2026.
- "Bug Triaging Based on Transformer Models Utilizing Commit Messages," ITNG 2025, Special Track Presenter, April 2025.

RESEARCH COMPUTING ALLOCATION

TACC Lonestar6 Allocation (NSF ACCESS, Project CCR25032) July 2025 – July 2026

- "Transformer-based Bug Triage using Hybrid Deep Learning Models": Lonestar6 at the Texas Advanced Computing Center. PI: Dr. Renee Bryce; I prepared the proposal documents and am the sole research user.

PROFESSIONAL EXPERIENCE

Senior Software Engineer (Android), Team Lead, HelloDoc24, Dhaka February 2021 – August 2022

- Led the Android team in designing and building a microcredit-based video consultancy application with role-based access control.
- Architected the app with MVVM, Kotlin Coroutines and Flow, Retrofit, and ViewModel (minimum SDK 21); integrated SignalR, Firebase Cloud Messaging, Azure Communication Services, WorkManager, Paging 3, and Jetpack.
- Implemented local and instrumented unit tests and automated UI tests (Espresso); managed production issues and root-cause analysis in an Agile process (Redmine, Git).

Software Engineer, Android Team Lead, Field Information Solutions Ltd., Dhaka January 2018 – January 2021

- Became team lead for three Android projects within the first year, including field-data collection apps for Mali Projets Eau, Stabilisation à Ménaka, and Pistes pour la Paix.
- Built reusable modules (configurable CAMERA module, unified single-login AUTH system, shared UTILITIES library) that made projects more modular and robust.
- Integrated automated testing and improved offline/online connectivity handling and legacy-system performance; contributed to BRAC Microfinance Rwanda, NKG Bloom Uganda, UNDP NUPRP, Win Win Myanmar, RIDF Remote Monitoring Tool, and Foodex Sales Tracker.

LEADERSHIP AND COMMUNITY ENGAGEMENT

General Secretary, Bangladesh Student Association, University of North Texas September 2024 – July 2025
Tech Lead, Team VeriWell, Innovate UNT 2025 Pitch Competition (finalist) May 2025
Team Lead, Smart City Hackathon, Bangladesh (Top 5) November 2016

HONORS AND AWARDS

Best Paper Award (Reviewers' Choice), IEEE CCWC 2026 January 2026
Excellent Program Committee Member, IEEE ICMLA 2025 December 2025
TACC Lonestar6 Compute Allocation (NSF ACCESS), Project CCR25032 July 2025
Finalist, Innovate UNT 2025 Pitch Competition (USD 5,000); tech lead, Team VeriWell May 2025
Special Track Presenter, ITNG 2025 April 2025
Student Travel Grant, UNT Toulouse Graduate School, College of Engineering, and CSE Department March 2025
International Education Scholarship, UNT International Affairs Fall 2024
CEO Prep Award, University of North Texas Fall 2024
Invited Attendee, Google I/O Connect, Miami, FL May 2023
Bangladesh–Sweden Trust Fund Travel Grant May 2023

Toulouse Graduate School Tuition Benefit, University of North Texas August 2022 – July 2025

PROFESSIONAL SERVICE

Program Committee Member, IEEE Int. Conf. on Tools with Artificial Intelligence (ICTAI 2026) 2026
Program Committee Member, IEEE Int. Conf. on Machine Learning and Applications (ICMLA 2026) 2026
Program Committee Member, IEEE ICMLA 2025 (recognized as Excellent Program Committee Member) 2025
Reviewer, IEEE ICMLA 2024 2024

CERTIFICATIONS

Google Certified Associate Android Developer 2017
Java Certificate, LICT Program / Ernst & Young 2016

TECHNICAL SKILLS

- **Programming:** Python, Java, C++, Kotlin
- **Machine learning and NLP:** transformer models (RoBERTa, CodeBERT, DeBERTa, XLNet, T5), deep learning, large language models, learning to rank (LightGBM LambdaMART), TensorFlow, scikit-learn, NLTK, Pandas, SageMaker
- **Teaching and collaboration tools:** Canvas LMS, Git and GitHub, Redmine, Agile/Scrum practices
- **Platforms:** TACC Lonestar6 (HPC), Azure, Android Studio, Visual Studio, Anaconda, WebRTC